DAY 34 | POST-READ

Observability: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Signals	Logs (events), metrics (aggregates), traces (timing per step), plus quality and safety signals.
Structured logs	One JSON object per line with request id, release, model, tokens, latency, retrieved ids, guardrail result.
Spans	Time each step inside a trace; nested spans show the call tree.
Dashboard	Traffic, latency, errors, cost, quality, safety.
Alerts	Sustained windows, relative changes, an owner and a runbook for each.
Drift kinds	Input, knowledge, model and behaviour drift.
Feedback loop	Collect, triage, fix, add a golden test, verify with the quality gate.
Feedback types	Explicit (thumbs, comments) and implicit (rephrasing, escalation, copying).

Part B: Going beyond today

1. OpenTelemetry

OpenTelemetry is the open standard for traces, metrics and logs. Instrument once and send data to Azure Monitor, Grafana, Datadog or others. There are emerging conventions for recording LLM calls in spans.

2. LLM observability tools

Tools such as Langfuse, LangSmith and Arize Phoenix record prompts, retrieved context, outputs, tokens and scores per request, and link them to evaluations. Check data handling policies before sending customer data to any tool.

3. Sampling

Logging full prompts for every request can be expensive and sensitive. Log metadata for all requests, and full content for a sample (with PII masked), plus every request with negative feedback.

4. Service level objectives

Agree targets with the customer, for example "95% of answers within 3 seconds" and "thumbs-down rate under 8%". Alerts then protect promises, not arbitrary numbers.

5. Privacy in logs

Mask personal data before logging, restrict who can read raw logs, and set retention periods that match the customer's policy and local law.

Part C: Check yourself

- What is the difference between monitoring and observability?
- Which fields should every LLM request log contain?
- Why use sustained windows in alert rules?
- Give an example of implicit feedback.
- What happens to a fixed failure in a good feedback loop?

Part D: Practice ideas for tonight

- Add JSON logs and spans to your capstone prototype.
- Write three alert rules with owners and runbook links for your capstone.
- Design the thumbs-down review screen: what does the reviewer need to see?

Coming up on Day 35: drift types and mitigation, quality optimisation and LLMOps automation.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today